Subject Description Form

Subject Code	ITC6983
Subject Title	Guided Study in Textiles & Clothing
Credit Value	3
Level	6
Pre-requisite / Co-requisite / Exclusion	Nil
Objectives	This subject provides a guidance for research students to carry out a systematic study of auxetic textiles and to explore applications of such non-conventional textile materials in functional clothing and technical areas.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Understand basic concepts of auxetic materials and their nonconventional behaviors b. Catch the up-to-date information about different types of auxetic materials and their applications c. Apply different auxetic geometries to develop auxetic materials by using textile technology d. Explore applications of auxetic textiles in different areas
Subject Synopsis/ Indicative Syllabus	**1. Basic concepts of auxetic materials** Definition of Poisson's ratio, effect of Poisson's ration on materials properties, auxetic and non-auxetic behaviors. **2. Auxetic geometries** Reentrant geometries, rotating units, folded structures, chiral structures, star honeycomb structures. **3. Auxetic fibers and yarns** Auxetic fibers formed based on special macromolecular structure and special spinning process, auxetic double helical yarns, auxetic composite yarns. **4. Auxetic fabrics** Methods used to produce auxetic fabrics, auxetic knitted fabrics, auxetic woven fabrics, auxetic braided fabrics, auxetic non-woven fabrics. **5. Auxetic composites** Methods used to produce auxetic composites, auxetic laminated composites, auxetic textile composites. **6. Applications of auxetic textile materials** Applications of auxetic textiles in clothing and technical areas, such as high performance underwear, sportswear, protective clothing, medical care, etc.

Teaching/Learning Methodology	This is a guided study course and will have derived self-studying. The subject lecturer will suggest reading materials based on student's research area and discuss with the students. Students are required to attend lectures, read assigned research papers, and submit a written report on an assigned research topic.

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
		a	b	c	d		
1. Oral presentation	40%	✓	✓	✓	✓		
2. Written report	60%	✓	✓	✓	✓		
Total	100 %						

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Oral presentation provides an opportunity to interact between the students and the lecturer, so that the understanding of the subject matter can be ensured.

The written paper can assess students' critical thinking and scientific analysis of various methods used in auxetic textile research.

Student Study Effort Expected

Class contact:	
▪ Lecture/presentation/discussion	12 Hrs.
▪ Lab demonstration	2 Hrs.
Other student study effort:	
▪ Self-reading	50 Hrs.
▪ Assignment and preparation of report	50 Hrs.
Total student study effort	114 Hrs.

Reading List and References	**Book** Teik-Cheng Lim. *Auxetic Materials and Structures*. Springer, 2015. **Journals** 1. Science 2. Natural Materials 3. Advanced Materials 4. Advanced Functional Materials 5. Smart Materials and Structures 6. Applied Physics Letters 7. Textile Research Journal 8. Physical Status Solidi (b)